InferenceChain

A Decentralized Blockchain Native to AI Inference

Quality of Inference Consensus · Proof of Quality of Inference · Sharded BFT
Technical Whitepaper & Vision Document
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1. Abstract

InferenceChain is a novel blockchain protocol designed specifically for decentralised AI inference. Unlike general-purpose blockchains that treat computation as an opaque black box, InferenceChain introduces inference quality as a first-class consensus property. Validators do not merely agree on transaction order — they agree on the correctness of deep neural network (DNN) outputs.
Three novel primitives drive the protocol: **Quality of Inference Consensus (QoI)**, a PBFT-derived algorithm that uses cosine similarity between inference vectors to reach Byzantine-fault-tolerant agreement; **Proof of Quality of Inference (PoQI)**, an on-chain reputation system that quantifies each validator's historical accuracy and weights their future influence accordingly; and **Sharded BFT (S-BFT)**, a horizontal scaling layer that partitions the validator set into independent committees to achieve linear throughput growth without sacrificing BFT safety guarantees.
A working testnet implementation — including the full consensus engine, P2P gossip layer, on-chain state machine, model admission protocol, OOD-based quality assessment, reliability tracking with

anti-gaming mechanisms, and a live web dashboard — has been developed and is described in this document.
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2. Problem Statement

The rise of AI inference as a service has created a critical centralisation risk. Today, virtually all AI inference is performed by a handful of large cloud providers. Users have no way to verify that the model they requested actually ran, that the output has not been tampered with, or that the service provider is not selectively serving degraded results.
Existing blockchain-based approaches fall into two categories, both inadequate:
InferenceChain addresses all four limitations simultaneously. Consensus is reached on inference quality, verified by mathematical comparison (cosine similarity). Reputation is an on-chain quantity, updated deterministically by protocol rules. Admission requires cryptographic proof of model capability before a node may participate. In-distribution awareness (OOD scoring) and hidden challenges prevent gaming and reward domain expertise.
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3. Architecture Overview

InferenceChain separates concerns cleanly across four layers:
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4. Quality of Inference Consensus (QoI)

QoI is a three-phase Byzantine-fault-tolerant consensus protocol derived from PBFT, adapted to agree not on an arbitrary value but on the quality of a DNN inference result. The key insight is that correct DNN inference on the same image produces output probability vectors that are mathematically close — measurable by cosine similarity.
The protocol requires $n \geq 3f + 1$ validators to tolerate f Byzantine nodes.

Phase 1 — PRE-PREPARE

The primary (proposer) receives an INFERENCE_REQUEST containing an image hash. It fetches the image from the requesting client, runs it through its local DNN model, and broadcasts a PRE_PREPARE message containing its output probability vector (softmax over 10 CIFAR-10 classes).

Phase 2 — PREPARE

Each DNN validator independently runs the same image through its own local model. It computes the cosine similarity between its output vector and the primary's vector. If the similarity exceeds the threshold, it broadcasts a PREPARE message with its own vector. The primary collects PREPARE messages until it has 2f+1.

Phase 3 — COMMIT

Once 2f+1 PREPARE messages are collected, the primary broadcasts a COMMIT message. Each validator that has seen 2f+1 PREPAREs broadcasts its own COMMIT. When a node collects 2f+1 COMMITs, the round is finalised. A QoI block is appended to the chain containing the consensus result, all signatures, and reputation deltas.

Quality Scoring

After each round, each validator's output vector is compared pairwise against all others. Validators whose vectors lie within the cosine similarity threshold of the majority cluster are marked as honest and receive reputation increases proportional to their similarity score. Outliers (Byzantine or low-quality nodes) are penalised.

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5. Proof of Quality of Inference (PoQI)

PoQI is the on-chain reputation system that tracks each validator's cumulative inference quality. It is not a separate consensus protocol — it is a deterministic state transition that occurs at the end of every QoI round.

Reputation Parameters

Reputation serves two functions: it weights the proposer election (higher reputation → higher probability of being selected as primary), and it weights block reward distribution. A validator that consistently produces high-quality inference results earns disproportionately more INFER tokens over time — creating a strong economic incentive for model quality maintenance.

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6. Proof of Model (PoM) — Admission Protocol

Before a node may participate in QoI consensus, it must pass the Proof of Model admission protocol. PoM prevents Sybil attacks and ensures that every active DNN validator in the network actually possesses a model that meets minimum quality standards.

Admission Flow

The weights hash committed in Step 1 binds the node to a specific model. If a node later attempts to serve a different model, the hash mismatch is detectable on-chain by any verifier — providing cryptographic proof of model substitution fraud.

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7. Sharded BFT Architecture (S-BFT)

Standard PBFT has $O(n^2)$ message complexity — as the validator set grows, the communication overhead grows quadratically, making it impractical beyond ~100 nodes. S-BFT solves this by electing a small quorum of K nodes per inference request, rather than involving the full validator set.

The Scaling Problem

With N DNN nodes in the network, running QoI across all of them for every inference request is unscalable:
• $O(N^2)$ message complexity
• Latency that tracks the slowest node
• Most nodes having no familiarity with the requested model or dataset
S-BFT reduces the active committee to K = 10 nodes (configurable), achieving $O(K^2)$ message complexity independent of total network size.

Per-Request Quorum Selection

For each inference request, a quorum is elected as follows:
1. **Filter** eligible DNN validators by model name and dataset ID matching the request
2. **Compute seed** = SHA-256(block_hash || request_id) — unique per round, reproducible by any node
3. **Sample K nodes** from the filtered pool using the seed — without replacement, deterministically
4. **Sort** selected nodes by node_id for canonical ordering
Quorum assignment is verifiable by any network participant and unpredictable before the inference request arrives — preventing adversaries from pre-positioning nodes.

Quorum Parameters

Local BFT Parameters

Each quorum operates with its own fault-tolerance parameters, independent of the global chain f:
$$f = \lfloor (K - 1) / 3 \rfloor$$
$$\text{threshold} = 2f + 1$$
For K=10: f=3, threshold=7. For K=4: f=1, threshold=3.
Nodes not elected to the quorum stand by for one block slot — they do not participate and do not interfere. The committed QoI block arrives via P2P gossip and is ingested normally by all nodes.

Quorum Integrity at Commit

When a QoI block is sealed, the protocol validates that every commit signature originated from a node in the elected quorum. Signatures from outside the quorum are stripped — those nodes receive no reward. The elected quorum node IDs are embedded in the ConsensusOutcome and written to chain in the CONSENSUS_RESULT transaction.
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8. Block Structure & Transaction Types

Two Block Types

Transaction Taxonomy

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9. OOD-Aware Quality & Anti-Gaming Infrastructure (NEW)

9.1 Out-of-Distribution Scoring

Every admitted DNN node automatically calibrates an **Out-of-Distribution (OOD) scorer** the moment its **`NODE_ADMITTED`** transaction is committed. The scorer measures how *in-distribution* (familiar) a candidate image is for a given node's model.
Scoring Method: Hybrid Mahalanobis + Energy
• **Shared Encoder**: All nodes use frozen ViT-B/16 (torchvision.models.vit_b_16)
• **Class-Conditional Profiles**: Mahalanobis distance on penultimate-layer features
• **Knowledge Score K(x)**: Logistic fusion yielding $K(x) \in [0, 1]$
• **Query-Time Signing**: Optional Ed25519 signatures for authenticity

9.2 Multi-Factor Weighted Quorum Selection

The quorum incorporates three factors: **stake**, **knowledge**, and **reliability**.
Weighting Formula
$$w_i = \text{stake}_i \times K_i^{\gamma} \times R_i^{\beta}$$
where $\gamma = 1.5$ and $\beta = 1.0$ (tunable).
Deterministic Weighted Sampling
Using Efraimidis-Spirakis algorithm for fair, reproducible selection without replacement.

9.3 Node Reliability Tracking (Anti-Gaming)

Per-node **EMA-based reliability** with dual update pathways:
• **Regular QoI rounds** (α=0.2): Target = 0.6 + 0.6×QoI_score (honest) or 0.2 (Byzantine)
• **Challenge rounds** (α=0.45): Target = 0.65 + 0.55×K (honest) or 0.25 – 0.2×K (Byzantine, knowledge-weighted)
Reliability score $\in$ [0.05, 1.5], clamped to prevent extremes.
Per-node counters: rounds, honest_rounds, byzantine_rounds, challenge_rounds, challenge_penalties.

9.4 Deterministic Hidden Challenge Rounds

Challenge Selection at configurable rate (default 15%):
Challenge Records: Timestamp, request_id, image_hash, block_hash, per-node honesty, OOD/knowledge scores.
Challenge-Driven Updates: Knowledge-weighted reliability adjustments — high-knowledge dishonest nodes receive strong penalties.
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10. What Is Already Built

The following components are fully implemented and operational on the local testnet as of April 2026.
✓ Blockchain core with QoI & PoS blocks, Merkle roots, persistent SQLite storage
✓ ChainState with balances, stakes, reputations, registered nodes, reliability/challenge records
✓ S-BFT quorum selection with multi-factor weighting (stake × knowledge × reliability)
✓ QoI consensus engine (3-phase PBFT, view-change, timeout)
✓ PoS consensus for simple transactions
✓ Proof of Model admission protocol
✓ OOD profiling with shared ViT-B/16 encoder and Mahalanobis profiles
✓ Reliability tracking with dual EMA pathways
✓ Deterministic hidden challenges with challenge records
✓ Wallet & signature verification (secp256k1, BIP39)
✓ P2P gossip layer with WebSocket connections
✓ REST API (40+ endpoints) with full OpenAPI documentation
✓ Model registry (7 CIFAR-10 architectures)
✓ Frontend dashboard (React)
✓ Desktop application (PyInstaller + Electron)
✓ Test suite (37/37 QoI consensus tests passing)

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11. Tokenomics — INFER Token

The INFER token is the native currency. It serves three roles: **economic incentive** for inference quality, **security bond** making Byzantine behaviour costly, and **governance weight** for future votes.

Supply Parameters

Reward Distribution

Stake × Reputation Interaction

$$\text{reward\_share}(i) = \frac{\text{stake}(i) \times \text{reputation}(i)}{\sum_j (\text{stake}(j) \times \text{reputation}(j))}$$

Quality and capital are equally weighted — new nodes earn nothing until demonstrating quality, preventing pure capital dominance.

Emission Schedule

~6.3M INFER/year maximum theoretical rate at 10 INFER per 5-second block target. Remaining 90M supply: ~14 years to emit. Actual emission demand-driven.

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12. Vision & Roadmap

InferenceChain's vision: become the trust layer for AI inference — the protocol any application can use to prove that a specific AI model produced a specific output, verified and economically incentivised.

Phase 1 — Foundation (Current: v0.5)

✓ Core consensus, reputation, model admission
✓ Sharded BFT with multi-factor weighting
✓ OOD-aware scoring and reliability tracking
✓ Deterministic hidden challenges
✓ Full implementation and testnet

Phase 2 — Scaling & Security (Q3 2026)

■ Tag-driven challenge pools (domain taxonomy)
■ Output-space label canonicalization
■ Signature aggregation (BLS)
■ Peer discovery (Kademlia DHT)
■ Multi-model support
■ Light client protocol

Phase 3 — Ecosystem (Q1 2027)

■ Smart contracts (Turing-complete VM)
■ Cross-chain bridge (EVM)
■ Model marketplace
■ DAO governance
■ Mainnet launch
■ Python/JS SDK

The Broader Vision

As AI becomes consequential across healthcare, law, finance, autonomous systems — verifiable, auditable AI becomes necessity. InferenceChain provides cryptographic infrastructure for this audit trail: any inference result on-chain can be independently re-verified by anyone, at any time, with mathematical certainty.
End state: AI inference as transparent, economically incentivised, cryptographically auditable process — not a black box.
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13. Conclusion

InferenceChain introduces a fundamentally new class of blockchain — designed from first principles for AI inference. The three core innovations (QoI, PoQI, S-BFT) with OOD-aware weighting, reliability tracking, and hidden challenges solve trustless, verifiable, quality-assured DNN inference at scale.
Complete working implementation demonstrates protocol is not merely theoretical. Testnet runs QoI rounds, admits validators through PoM, tracks on-chain reputation and reliability, manages challenge rounds, and serves full-featured dashboard — all operating coherently.

INFER token creates self-sustaining economic system where quality and security incentives align: stake securing network proportionally determines reward shares; reputation — earned only through quality — amplifies stake's influence.

InferenceChain is positioned to become the trust infrastructure for the AI era — making AI outputs as verifiable as blockchain transactions.

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